

MODULE 3

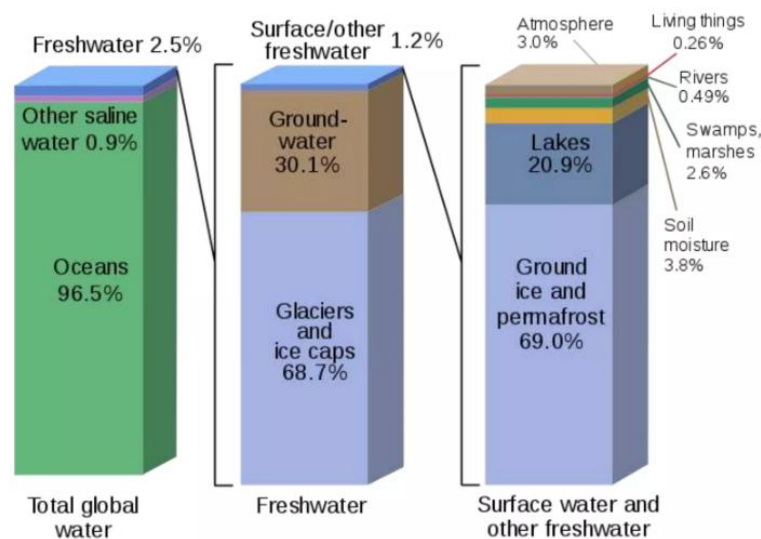
WATER

- A single water molecule is composed of two atoms of hydrogen attached to one atom of oxygen (H₂O) by covalent bonds
- About 71 percent of the Earth's surface is water-covered
- [Oceans](#) hold about 96.5 percent of all Earth's water
- Water also exists in the air as [water vapor](#), in [rivers](#) and [lakes](#), in icecaps and [glaciers](#), in the ground as soil moisture, in [aquifers](#), and even in you and other organisms.

SOURCES AND AVAILABILITY OF FRESHWATER

- Water resources are sources of water that are useful or potentially useful
- Uses of water include agricultural, industrial, household, recreational, and environmental activities
- Virtually all of these human uses require fresh water
- 97.5% of the water on the Earth is saltwater
- Only 2.5% is fresh water
- slightly over two-thirds of this is frozen in glaciers and polar ice caps
- The remaining unfrozen freshwater is found mainly as groundwater, with only a small fraction present above ground or in the air

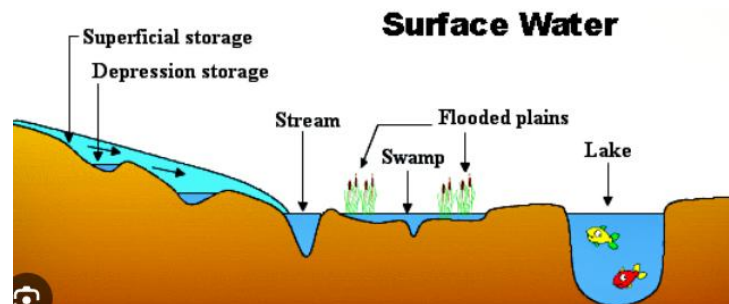
Where is Earth's Water?



SOURCES OF FRESH WATER

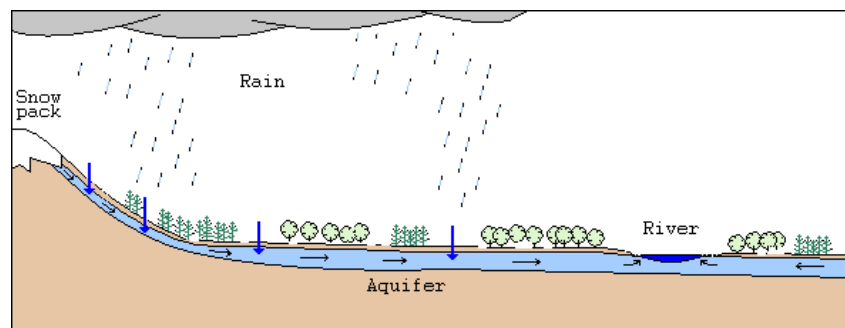
1. Surface Water

- Surface water is water in a river, lake or fresh water wetland
- Surface water is naturally replenished by:
 - Precipitation and naturally lost through discharge to the oceans
 - Groundwater recharge



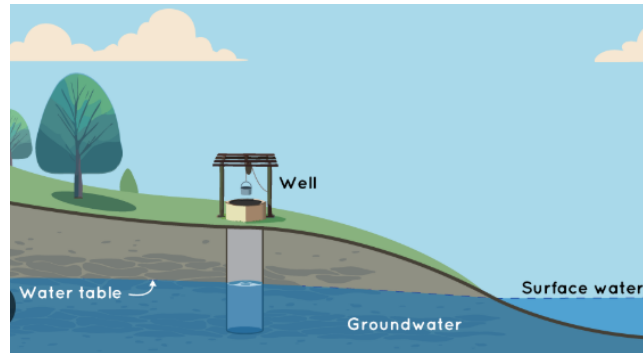
2. Under River Flow

- Under river flow is the water that flows through the rocks and sediments below a river. This flow is often called the hyporheic zone.
- The hyporheic zone is the area below a river where water flows through rocks and sediments.
- The hyporheic zone can be a significant part of a river's flow, especially in large valleys.
- The hyporheic zone can exchange water with aquifers, which are large underground areas that store water.



3. Groundwater

- Groundwater is fresh water located in the subsurface pore space of soil and rocks
- It is also water that is flowing within aquifers below the water table
- Sometimes it is useful to make a distinction between groundwater that is closely associated with surface water and deep groundwater in an aquifer (sometimes called "fossil water").



4. Frozen Water

- Frozen water, primarily in the form of glaciers and ice caps, holds about 68% of the Earth's fresh water
- As climate change accelerates ice melt, this frozen water becomes a significant source of fresh water for rivers and lakes
- In regions like the Himalayas and the Andes, glacial meltwater sustains agriculture and drinking supplies for millions.
- However, excessive melting can lead to water scarcity in the long term as glaciers shrink.

5. Desalination

- Desalination is an artificial process by which saline water (generally sea water) is converted to fresh water
- The most common desalination processes are distillation and reverse osmosis.
- Desalination is currently expensive compared to most alternative sources of water, and only a very small fraction of total human use is satisfied by desalination

WATER CONSERVATION

The process of saving water for future utilization is called water conservation. With the increase in population, the quality & quantity of water have declined.

Water conservation strategies

1. Reducing evaporation

- Evaporation of water is commonly found in humid regions.
- It can be reduced by placing horizontal barriers of asphalt below the soil surface, which increases water availability and crop yield.

2. Reducing irrigation loss

- Water loss during irrigation can be reduced by the following methods.
 - a. Growing hybrid crop varieties
 - b. Irrigation in early morning or later evening

- c. Using drip irrigation or sprinkler irrigation

3. Re-use of water

- Treated water can be used for fertile-irrigation
- Grey water can be used for washing cars, watering gardens, etc.
- Water used for washing vegetables and fruits can be used to water plants.



4. Preventing wastage of water

- Wastage of water can be prevented by:
 - Closing the taps when not in use
 - Repair any leakage from pipes
 - Use small-capacity taps

5. Decrease runoff loss

- Run-off can be reduced by allowing water to infiltrate into the soil
- This can be done by contour cultivation or terrace farming

6. Avoid discharge of sewage

- Discharge of untreated sewage into natural water resources must be prevented

METHODS OF WATER CONSERVATION

There are many methods available for conservation of water, some of the important measures are:

1. Rainwater harvesting

Rainwater harvesting is a technique of collecting and storing rainwater in natural reservoirs or tanks for future utilization

Objecting to rainwater harvesting:

- To meet the increasing demands of water
- To reduce groundwater contamination

- c. To minimize the water crisis
- d. To reduce soil erosion
- e. To raise the water table by recharging the groundwater

ROOFTOP RAINWATER HARVESTING

- It is a method of collecting rainwater from the roof of the building and storing it for future use
- It is a low cost and effective technique for urban buildings



ADVANTAGES OF RAINWATER HARVESTING

- Improves the quality and quantity of groundwater
- Reduces soil erosion and flooding
- Increases the availability of water
- Promotes water and energy conservation

2. Watershed management

Watershed

Land area from which water drains under the influence of gravity into a stream, lake, reservoir or other body of surface water.

Watershed management

The management of rainfall and resultant runoff is called watershed management.



Factors affecting the watershed:

1. Overgrazing

Overgrazing occurs when livestock feed excessively on vegetation. It reduces plant cover, exposes soil to erosion, decreases infiltration, and increases surface runoff. This leads to soil degradation and sedimentation in water bodies within the watershed.

2. Deforestation

Removal of trees and forest cover reduces interception and infiltration of rainfall. It increases soil erosion, surface runoff, and flood risk. Deforestation also disturbs the ecological balance and reduces groundwater recharge in the watershed.

3. Mining

Mining activities disturb the land surface and remove vegetation. They produce large quantities of waste materials and pollutants, which may enter streams and rivers. This causes soil erosion, water contamination, and degradation of watershed quality.

4. Construction activities

Urban development, road construction, and building activities increase impervious surfaces like concrete and asphalt. This reduces groundwater recharge, increases runoff, causes flooding, and leads to sediment and pollutant discharge into water bodies.

5. Droughty climates

In regions with low rainfall, vegetation cover is scattered and soil moisture is limited. This reduces groundwater recharge and stream flow. Prolonged drought conditions can degrade soil structure and reduce the overall productivity of the watershed.

Objectives of watershed management

- To minimize the risks of floods, droughts, and landslides
- To prevent soil erosion by runoff
- To raise the groundwater level
- To generate employment opportunities
- To develop rural areas in the region

Watershed management techniques

1. **Trenches:** Trenches are dug at equal intervals to improve groundwater storage

2. **Earthen dam:** An earthen dam is constructed in the catchment area to check the water run-off
3. **Farm pond:** A farm pond can be built to improve the water storage capacity of the catchment area
3. **Dykes:** Dykes must be built along nullahs to raise the water table



Trenches



Earthen Dam



Farm Ponds



Dykes

WATER POLLUTION

- Water pollution can be defined as the presence of **some foreign substances or impurities** in water in such a quantity to constitute a health hazard, thus making water unfit for use.
- Water contained in water bodies like lakes, rivers, and oceans is called surface water
- Water stored in aquifers (underground rock structures) is called underground water (subsurface water)
- Both surface and underground surface (subsurface water) are prone to pollution

The Water (Prevention and Control of pollution) Act, 1974

- The Water Act, 1974 is the first Indian law focused on the **prevention of discharging domestic and industrial wastewater into water bodies without treatment.**

- This Act provides for the establishment of pollution control boards at Centre and States to act as watchdogs for prevention and control of pollution.
- The Act was passed by the Parliament under Article 252 of constitution of India and was amended in 1978 and 1988.

Objectives of the Water Act

- Prevention and control of water pollution
- Maintaining or restoring the wholesomeness of water
- Establishment of boards of the prevention and control of water pollution

According to the Water Act, 'pollution' is defined as contamination of water or alteration of the physical, chemical or biological properties of water, or the discharge of any sewage or trade effluent (directly or indirectly) which is likely to render such water harmful or injurious to:

- Public health or safety
- Domestic, commercial, industrial, agricultural or other uses
- Life and health of plants, animals or aquatic organism

Water pollution caused by these sources can be minimized if the effluents from these sources are controlled, treated up to acceptable levels and disposed of.

GROUNDWATER POLLUTION

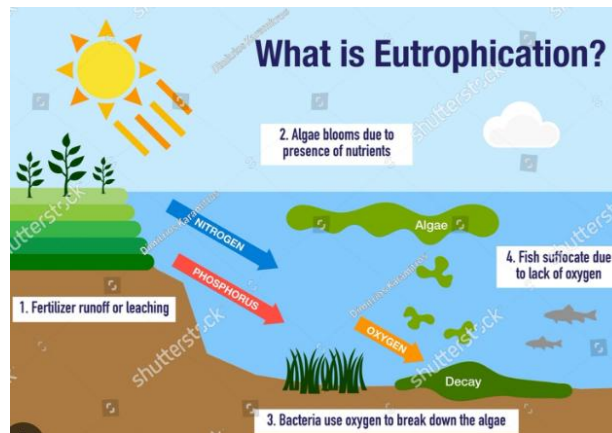
- Water pollution is not confined to surface water alone.
- Various kinds of harmful materials like fertilizers, pesticides, metals, etc present in solid waste (disposed on land) get dissolved or leaked into water.
- During rain these pollutants present in water percolate down into the soil and contaminates the groundwater beneath.

WATER POLLUTANTS

Water pollutants are substances that cause water pollution. Some of the common pollutants in water are:

1) Inorganic salts

These include nitrates, phosphates, carbonate chlorides, sulphates of Ca, Mg, Fe etc. These salts make water "hard". Hard water increases soap consumption and deposits scales on pipelines. Iron causes spots and stains on white fabrics. Water bodies enriched with nitrates and phosphates induce rapid growth of microscopic plant life which can lead to algal bloom popularly known as Eutrophication.



2) Acids/Alkalis

Most freshwater bodies have a natural pH in the range of 6 to 8. Acids and alkalis are discharged into streams by chemical industries and other industrial plants. High concentrations of acid, sufficient to lower the pH to below 7.0 causes eye irritation to swimmers, rapid corrosion of ship's hull, and deterioration of fishermen's nets. When the pH of water goes below 5, fish populations begin to disappear. Also, the efficiency of water-treatment plants is affected by acidity. Alkalies (e.g. NaOH) appear in wastewater from soap manufacturing, textile dyeing, rubber reclamation, leather tanning, etc. High pH is also detrimental to aquatic life.

3) Organic matter

Organic pollution occurs when large quantities of organic compounds, which act as substrates (food) for microorganisms, are released into water sources. During the decomposition of organic pollutants, the dissolved oxygen in the receiving water may be used up at a greater rate than it can be replenished, causing oxygen depletion. The minimum amount of dissolved oxygen in water for the sustenance of fish life is 3 to 4 mg/L.

4) Suspended solids

Suspended solids either settle to the bottom or remain suspended in the water body. These solids increase the turbidity of the water body. Solids that get deposited on the banks decompose and cause odour nuisance.

5) Floating matter

These include oils, greases and other materials which float on the surface, and not only make the river unsightly, but also obstruct passage of light through the water. Presence of floating matter hinders the self-purification process of water bodies as well.

6) Thermal discharges

Increase in the normal temperatures of natural waters is caused by nuclear power plant/boiler/industrial discharges (where water is used as a cooling agent). Since warm water is lighter than cold, stratification occurs in the water body, and this causes most of the aquatic life to retreat to the stream bottom. Also, bacterial action increases, which in turn results in accelerated depletion of the stream's dissolved oxygen content.

7) Colouring materials

Colour in water is an indicator of pollution (which often invites public attention). Colour interferes with the transmission of sunlight into the stream, thus reducing its natural disinfection action. Colour in water is contributed by the effluents discharged from textile industries, paper mills, tanneries, slaughterhouses, etc.

8) Toxic chemicals

Highly complex, both organic and inorganic toxic compounds produced by various chemical industries have proved extremely toxic to aquatic life. These include cyanides, sulphide, acetylene, alcohol, etc. Many of these have a cumulative effect on the aquatic environment. For example, hexavalent Chromium found in wet cement is water-soluble and can soak into skin and produce an allergic reaction.

9) Micro-organisms

The domestic sector, food processing industries, and slaughterhouses usually discharge wastes containing microorganisms. Diarrhoea and Pneumonia are the biggest killer diseases in children aged under five years in India. Oral Rehydration Salts (ORS) with zinc supplementation is the correct treatment for Diarrhoea (as per 2013 WHO recommendation). The above micro-organisms, mainly bacteria, are of two types: a) Micro-organisms which assist in the degradation of the organic matter, b) Micro-organisms which are pathogenic in nature.

10) Radioactive materials

Radioactive waste typically comprises many radio isotopes (which are unstable) emitting ionizing radiation, which can be harmful to humans and the environment. This radiation is not readily detectable by the methods usually employed to determine the presence of contaminants in the environment. Exposure to high levels of radioactive waste may cause mutation and cancer.

11) Foam-producing matter

Presence of foam-producing matter leads to an undesirable appearance of the receiving streams /water bodies, which is usually discharged by textile mills, pulp and paper mills and some chemical manufacturing industries.

Some of the noticeable signs of water pollution are:

- Offensive odour from rivers, streams, lakes and oceans
- Oily and greasy material floating on the surfaces of water bodies
- Unchecked growth of aquatic weeds in water bodies
- Bad taste of drinking water
- Decrease of aquatic life in fresh water bodies

Impacts of water pollution

Water pollution can cause a variety of problems to human beings and other living things; some of them are listed below:

1. Consuming contaminated water leads to many health problems due to water-borne diseases (cholera, typhoid, jaundice, etc).
 - Excess amount of metals in drinking water (fluoride, lead, copper, etc) can cause general body weakness, mental retardation, failure of internal organs, cancer and even death.
2. Pollutants in water lead to widespread destruction of aquatic life
3. Nutrient-rich (nitrogen and phosphate) wastewater leads to abundant growth of algae, which results in eutrophication
4. Toxic pollutants in water, easily enter the food chain, which results in bio-magnification [Bio-magnification is the increase in the concentration of a pollutant from one link to another in a food chain].

CONTROL MEASURES OF WATER POLLUTION

1. Drinking water should be boiled, cooled, and then used
2. Disinfection of drinking water should be done by using chemicals like bleaching powder
3. Pesticides and insecticides should be prevented from nearby use of water lakes, ponds, and pools.
4. Drainage water should not be allowed to mix with drinking water
5. The drainage system should be maintained properly
6. The chlorination process is to be adopted for drinking water.
7. Appropriate wastewater treatment is to be provided before discharging wastewater from industries and domestic sectors
8. Using phosphate-free detergents and reducing the use of fertilizers (in agriculture), the phenomenon of eutrophication can be controlled
9. The public should be made aware of the ill-effects of disposing of wastewater onto waterbodies
10. Legislative measures have to be made more stringent and the principle of PPP (Polluter Pays Principle) should be strictly followed [i.e. polluter should bear the cost of cleanup operations, since polluter is responsible for pollution]

WATER QUALITY STANDARDS – (Also **Refer (IS 10500:2012))**

Characteristics	Parameters	Acceptable Limit	Permissible Limit
1. Physical	Colour	5 Hazen	15 Hazen
	Turbidity	1 NTU	5 NTU
	Taste	Agreeable	Agreeable
	Odour	Agreeable	Agreeable
	Total Dissolved Solids	500 mg/L	2000 mg/L
2. Chemical	pH	6.5-8.5	No relaxation
	Alkalinity	200 mg/L	600 mg/L
	Hardness	200 mg/L	600 mg/L
	Chloride (Cl)	250 mg/L	1000 mg/L
	Nitrate (NO ⁻³)	45 mg/L	No relaxation
	Iron (Fe)	0.3 mg/L	No relaxation
	Chromium (Cr)	0.05 mg/L	No relaxation

Characteristics	Parameters	Requirements
3. Biological	E. Coli or thermotolerant coliform bacteria	Shall not be detectable in any 100 ml sample

Water Quality Index (WQI)

- WQI indicates the quality of water in terms of an index number.
- It represents the overall quality of water for any intended use.

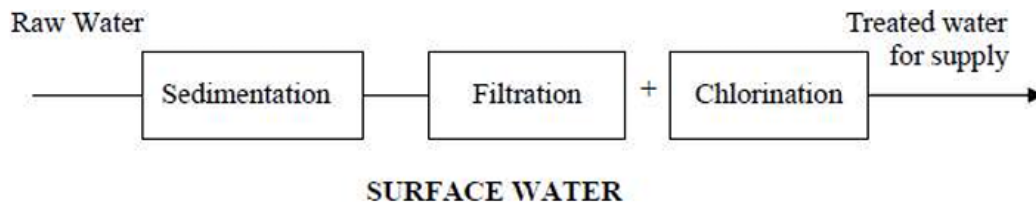
- It is defined as a rating reflecting the composite influence of different water qualities. parameters were taken into consideration for the calculation of the Water Quality index (WQI).
- The indices are among the most effective ways to communicate the information on water quality trends to the general public or to the policy makers, and in water quality management.
- In the formulation of the water quality index, the relative importance of various parameters depends on the intended use of water.
- Mostly it is done from the point of view of its suitability for human consumption.

WQI	Water quality
< 25	Excellent
26-50	Good
51-75	Fair
76-100	Poor
101-150	Very Poor
> 151	Worst

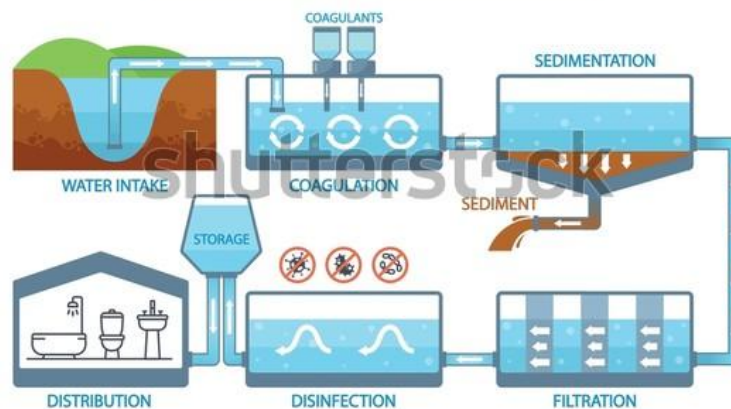
WATER TREATMENT

- Water treatment is the process of removing contaminants from wastewater and household water.
- It includes physical, chemical, and biological processes to remove physical, chemical, and biological contaminants.
- Its objective is to produce an environmentally safe fluid waste stream and a solid waste suitable for disposal or reuse.

OVERVIEW OF WATER TREATMENT PLANT



WATER PURIFICATION



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Water Treatment Methods

Water Treatment methods include:

- Screening
- Aeration
- Plain sedimentation
- Sedimentation aided with coagulation
- Filtration
- Disinfection
- Softening

Name of unit	Purpose
Screening	Used for the filtration of coarser floating and suspended materials
Aeration	Used for removing dissolved gases like carbon dioxide, some taste and odour producing compounds like methane, hydrogen sulphide, the volatile organic compounds, removal of iron and manganese

Plain sedimentation	Used for removing the suspended particles from the water that could not be removed in the screening process
Sedimentation aided with coagulation	Used for removing fine clay and colloidal impurities which are electrically charged and are continually in motion and never settle down due to gravitational force
Filtration	Used for removing bacteria, colour, taste, and odours
Disinfection	Used for removing pathogenic bacteria
Softening	Used for removing hardness in water

1. SCREENING

Screens are fixed in the intake works or at the entrance of the treatment plant to filter coarser floating and suspended materials. So, the larger materials like papers floating in the water, wood, leaves, aquatic plants, polyethene, these kinds of things may come floating with the water, and then they need to be filtered out so that it does not enter into the main basin or through the equipment and pump, and do not spoil them. The screens are placed at an angle of 30° to 60° with the direction of flow.



2. AERATION

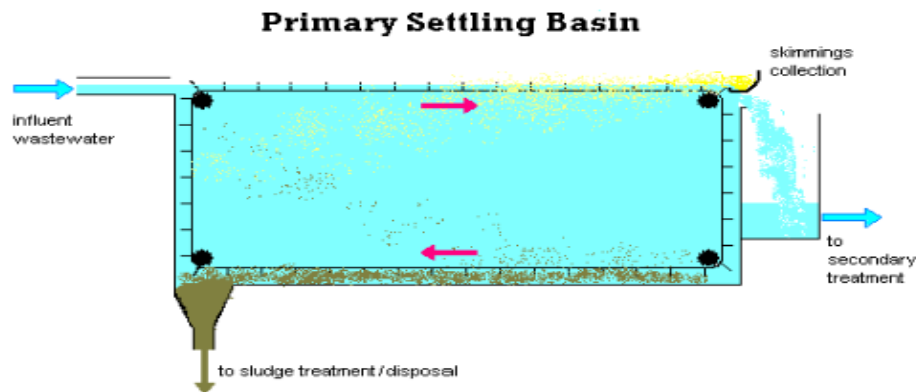
Aeration is usually effective against several pollutants, are discussed the dissolved gases like random carbon dioxide, some taste and odour producing compounds like methane, hydrogen sulphide, and volatile organic compounds like MTBE (Methyl Tertiary-Butyl Ether) or industrial solvents. It can also raise the pH of water, and it can be used for precipitation and removal of iron and manganese.

3. SEDIMENTATION

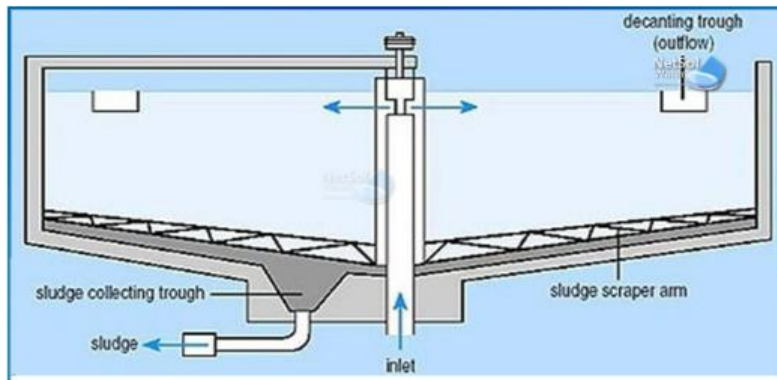
Sedimentation is the process that removes the suspended particles from the water that could not be removed in the screening process. The following are the types of settling tanks based on the shape:

- Rectangular tanks with horizontal flow

- Circular tanks with radial or spiral flow



Rectangular Sedimentation Tank



Circular sedimentation tank

4. SEDIMENTATION AIDED WITH COAGULATION

When water contains fine clay and colloidal impurities that are electrically charged, they are continually in motion and never settle down due to gravitational force. Certain chemicals known as coagulants are added to the water to remove impurities that are not removed by plain sedimentation. The chemical form insoluble, gelatinous, flocculent precipitate absorbs and entangles very fine suspended matter and colloidal impurities during its formation and descent through water. These coagulants further have the advantage of removing colour, odour and taste from the water.

5. FILTRATION

The process of passing the water through beds of sand or other granular materials is known as filtration. Filters are used for removing bacteria, colour, taste, odours, and producing clear and sparkling water. By sand filtration, 95 to 98% suspended impurities are removed. Filtration is carried out in three types of filters: Slow sand filter, Rapid sand filter, and Pressure filter.

6. DISINFECTION

The filtered water may normally contain some harmful disease-producing bacteria in it. These bacteria must be killed in order to make the water safe for drinking. The process of killing these bacteria is known as Disinfection.

PHYSICAL METHODS

- **Boiling:** Boiling the water for 15 to 20 minutes kills the disease-causing bacteria. This process applies to individual homes. The bacteria present in water can be destroyed by boiling it for a long time. However, it is not practically possible to boil huge amounts of water. Moreover, it cannot take care of possible future contaminations.
- **Ultra-Violet Rays:** Water is allowed to pass about 10cm in thickness by ultraviolet rays. This process is very costly and not used at water works. Suitable for institutions.

CHEMICAL METHODS

- **Chlorination:** Using chlorine gas or chlorine compounds. The germicidal action of chlorine, according to which the chlorine enters the cell walls of bacteria and kills the enzymes that are essential for the metabolic processes of living organisms.
- **Treatment with Excess Lime:** Lime is used in a water treatment plant for softening. But if excess lime is added to the water, it can also kill the bacteria. Lime when added, raises the pH value of water, making it extremely alkaline. This extreme alkalinity has been found to be detrimental to the survival of bacteria. This method needs the removal of excess lime from the water before it can be supplied to the general public. Treatment like recarbonation for lime removal should be used after disinfection.
- **Bromine and iodine:** Bromine and iodine are halogen-based, powerful disinfectants. Bromine is commonly used to control algae and bacteria in swimming pools, hot tubs, and industrial cooling towers. Iodine is primarily used for emergency water purification, in traveller's water kits, and as a medical antiseptic for skin/wounds. It is expensive and leaves taste and odour
- **Potassium permanganate:** This method is used for disinfection of dug well water, pond water or private source of water.
- **Ozone:** Ozone readily breaks down into normal oxygen, and releases nascent oxygen. The nascent oxygen is a powerful oxidising agent and removes the organic matter as well as the bacteria from the water. Very expensive process, leaves no taste, odour or residual

SUSTAINABLE WATER USE

Sustainable water use is the practice of managing water resources in a way that meets current and future needs without harming the environment. It involves reducing consumption, reusing water, and using new technologies.

Principles of Sustainable Water Use

- **Efficiency:** Maximizing water use with minimal waste

- **Equity:** Ensuring that all sectors (domestic, agriculture, industry) have fair access to water resources
- **Environmental Protection:** Maintaining healthy ecosystems that depend on water, such as wetlands, rivers, and aquifers
- **Resilience:** Ensuring that water resources can withstand and recover from stresses such as droughts, floods, and pollution.

Importance of Sustainable Water Use

- **Resource Scarcity:** With growing water demand, especially in arid and semi-arid regions, sustainable management ensures continued access to freshwater
- **Climate Change:** Unpredictable weather patterns increase the need for adaptive water management
- **Public Health:** Clean water is essential for health, and sustainable practices help maintain water quality.

CONFLICTS OVER-WATER RESOURCE

Conflicts over water resources are disputes, tensions, or violent clashes between countries, states, or groups regarding access to, control over, or usage of water. Driven by scarcity, high demand, and pollution, these conflicts often involve sharing transboundary rivers or managing infrastructure like dams.

Reasons for conflicts over-water resources

- **Competition for Limited Resources:** When water is scarce or unevenly distributed, competition between sectors (agriculture, industry, domestic) increases
- **Transboundary Water Issues:** Shared rivers and aquifers often lead to conflicts between countries or regions over water rights and usage
- **Pollution:** Contamination of water sources can lead to disputes over water quality and access
- **Climate Change:** Changes in rainfall patterns and rising temperatures exacerbate water scarcity, leading to conflicts over access to available water resources

Impact of Water Conflicts

- **Social and Political Instability:** Conflicts over water can lead to regional tensions and even military conflicts
- **Economic Consequences:** Water scarcity can lead to reduced agricultural yields, industrial output, and power generation, impacting economies
- **Environmental Degradation:** Over-extraction of water can harm ecosystems, causing river drying, habitat loss, and biodiversity reduction

Global Case Studies

Nile River Conflicts

Ethiopia - Background

- Treaties regarding Nile River water rights date back to the colonial period and continue to be referenced today in legal matters.
- In the colonial period of the early 20th century, historically significant treaties were signed between colonial powers and Nile River riparian states, including the Treaty Between Great Britain and Ethiopia, the Tripartite Treaty, and the Agreement between Egypt and Anglo-Egyptian Sudan.
- The treaties significantly undermined Ethiopia's rights to the Nile River, despite the Blue Nile running through Ethiopia's sovereign land, and favored Egypt and Sudan's rights to the Nile's water.

Egypt and Sudan - Background

- Both Egypt and Sudan are Lower Nile Basin countries and receive relatively low rainfall levels compared to Upper Nile Basin states, making them highly dependent on the Nile River for water supply.
- According to the UN, Egypt has an annual water deficit of around seven billion cubic meters and is projected to be completely water-scarce by 2025.
- Egypt's water scarcity resulted in protests in 2007, known as the "Revolution of the Thirsty," after residents of the Nile Delta reported putrid water in their limited water supply.
- As such, both countries continue to reference these colonial-era treaties in a bid to remain water-secure despite environmental pressures. However, Ethiopia refuses to recognize the treaty's validity due to its colonial context and unadulterated bias against Ethiopian water rights.
- Tensions in 1978 following Ethiopia's proposal of dam construction on the Blue Nile.
- Ethiopia's proposal was met with significant Egyptian backlash that ultimately led to the failure of the project.
- In 2010, Ethiopia announced its intention to build the largest and most expensive hydroelectric dam in Africa. The proposal was immediately met with Egyptian and Sudanese backlash, with Egypt calling for African Union (AU) and United Nations (UN) intervention on the matter and then Egyptian Minister of Water and Environmental Resources, Mohamed Nasr Eldin Allam, forming an agreement with the Sudanese government to preserve "the historical rights of both countries to the Nile's water."
- Although Sudan declared neutrality in the dispute in 2014, negotiations between all three states continue today to no avail and tensions continue today with the construction of the Ethiopian Grand Renaissance Dam.

The Colorado River

- United States and Mexico over water rights to the Colorado River.

- The Colorado River provides water to more than 40 million people across seven southwestern states, 29 tribal nations and Mexico
- In Las Vegas, 90% of its water supply comes from the river. In Phoenix and was it's 50% and in Los Angeles it's 25%.
- According to a 1922 agreement, each of these seven states have a legal right to a certain amount of the river's water. But this compact was made under the assumption that there was more water than there actually was.
- Scientists have been predicting for years that the Colorado River would continue to deplete due to global warming and increased water demands
- Between January 2000 and April 2023, the amount of water stored in Lake Mead and Lake Powell, the two largest reservoirs in the United States, declined by 33.5 million acre feet (41.3 billion cubic meters).

Interstate River disputes in India

- Krishna water dispute between states of Maharashtra, Karnataka and Andhra Pradesh (old)
- Godavari water dispute between states of Maharashtra, Andhra Pradesh, Karnataka, Madhya Pradesh and Goa
- Narmada water dispute between the states of Rajasthan, Madhya Pradesh, Gujarat and Maharashtra
- Cauvery water dispute between the states of Karnataka, Kerala, Tamil Nadu and Union Territory of Pondicherry
- Mahadayi/Mandovi water dispute between the states of Goa, Karnataka and Maharashtra

WASTEWATER

Wastewater or sewage is the byproduct of many water uses. There are household uses such as showering, dishwashing, laundry, and flushing the toilet. Additionally, companies use water for many purposes, including processes, products, and cleaning or rinsing of parts. After the water has been used, it enters the wastewater stream, and it flows to the wastewater treatment plant. These wastewater plants are complex facilities and provide a high-quality end product

Need for wastewater treatment

The need for wastewater treatment is to remove the wastewater pollutants to protect the environment and public health. When water is used by our society, the water becomes contaminated with pollutants. If left untreated, these pollutants would negatively affect our water environment. For example, organic matter can cause oxygen depletion in lakes, rivers, and streams. This biological decomposition of organics could result in fish kills and/or foul odors. Waterborne diseases are also eliminated through proper wastewater treatment.

Additionally, there are many pollutants that could exhibit toxic effects on aquatic life and the public. The necessity of wastewater treatment includes:

- The wastewater treatment plant plays an important role for mankind.
- The treatment of wastewater has become the need of the hour as it stops spreading diseases and illnesses caused by the wastewater.
- It helps society in making the water as well as the environment clean.

The treatment plant works are composed of three stages. The three stages of these plants include the primary stage, the secondary stage, and the tertiary stage.

- In the **primary stage/preliminary**, the contaminants that are easy to eliminate are taken out from the wastewater. These substances may include oils, grease, and fats that can be easily removed from the surface area. The solid things like grits, stones, rocks, etc., are strained.
- At the **secondary stage**, the removal of biological contaminants in wastewater takes place.
- At the **tertiary treatment**, which is the last stage of the plant, the water is cleaned purely to be discharged into the environment. This is composed of man-made or artificial systems that help in filtration. At this stage, the nitrogen and phosphorus content is eliminated from the water. In addition to this, the water is further disinfected using chemicals like chlorine as well as UV treatment.

Through these stages, the final water that comes out is clean and free from pollutants that can be safely released to the environment.

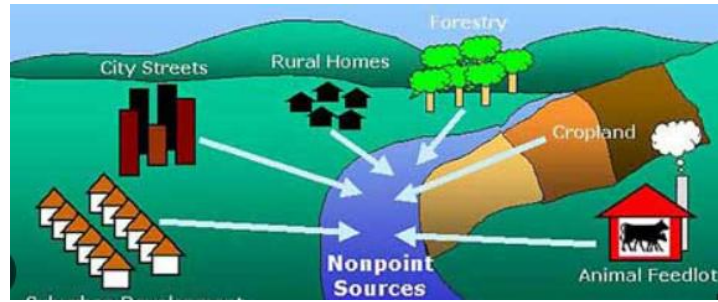
SOURCES OF WATER POLLUTION

1. Point Source

- Sources which can be readily identified at a single location (direct discharges from a single point)
- Some of the examples are wastewater discharge from industries, wastewater discharge from the domestic sector, etc.

2. Non-point Source or Diffused Sources

- Non-point sources are diffused across a broad area
- Their source of origin cannot be traced to a single discharge point (whose location cannot be easily identified)
- Examples include run off from agriculture lands, mining area etc.
- It is very difficult to control pollution from non- point sources



Natural Sources of Water Pollution

- Rainwater
- Atmosphere
- Underground rocks
- Volcanoes

Anthropogenic Sources of Water Pollution

- Oil spills
- Industrial wastewater discharges
- Solid-waste disposal
- Runoff from agricultural fields
- Wastewater from automobile garages, etc

In India, water pollution comes mainly from the following three sources:

- Domestic wastewater
- Agricultural Runoff
- Industrial Effluent

1. Domestic sewage

- Domestic sewage is the wastewater generated from the household activities (residential community) that originates from residential areas, commercial places, institutions and other public places
- Generally domestic sewage consists of 99.9% of water and 0.1% of solids
- These solids are mostly organic (which is extremely putrescible) with a small fraction of inorganic matter
- Its decomposition produces large quantities of malodorous gases and contains a number of pathogens (harmful micro-organisms)
- Domestic wastewater can be divided into two categories:
- **Black water (wastewater containing human excreta and urine)**
- **Grey water (wastewater generated from the bathroom and kitchen)**

2. Agricultural Wastewater

- Agricultural wastewater is the run-off from the agricultural fields and animal farms
- This wastewater is considerably rich in Nitrogen, Phosphate, Organic matter and Pesticides.

- Water bodies enriched with nutrients (Nitrogen and Phosphate) induce rapid growth of microscopic plant life (algae) in surface waters which lead to algal bloom thus reducing the oxygen content in the aquatic environment popularly known as Eutrophication (which is most common in stagnant water bodies such as ponds and lakes).

3. Industrial Wastewater

- Industrial wastewater (effluents) is the one which results from industrial operations
- Every year, the world generates around 400 billion tons of industrial waste, much of which is discharged into rivers, oceans and other waterways.
- Industrial wastewater may have pollutants of almost all kinds ranging from simple nutrients and organic matter to complex toxic substances
- The nature and composition (characteristics) of the industrial wastewater vary widely from industry to industry, and even for the same industry depending upon: Raw materials, Processes & Operational factors.

WASTEWATER QUALITY

1. Physical characteristics

Characteristics	Fresh sewage	Septic sewage
Temperature	Approximately 25-30°C	>30°C
Turbidity	Less turbid	More turbid
Odour	Musty smell	Foul smell
Colour	Lighter shade (Yellowish, Grey or light brown)	Darker shade (Black or dark brown)

2. Chemical characteristics

1. Total solids, suspended solids, and settleable solids

- Suspended solids - remain floating in sewage
- Dissolved solids - dissolved in sewage just as salt in water
- Colloidal solids are finely divided solids remaining either in solution or in suspension
- Settleable solids are that portion of solid matter which settles out, if sewage is allowed to remain undisturbed for a period of 2 hours

2. pH value

- If the pH value is less than 7, the sewage is acidic
- If the pH value is more than 7, the sewage is alkaline
- The pH value can be measured – potentiometer

- Fresh sewage is alkaline in nature
- Septic sewage is acidic in nature

3. Chloride content

- Chlorides are generally found present in municipal sewage
- It is derived from the kitchen wastes, human feces, and urinary discharges, etc
- The chloride content can be measured by titrating the wastewater (i.e. sewage) with as indicator, as is done for testing water supplies

4. Nitrogen content

- The presence of nitrogen in sewage indicates the presence of organic matter, and may occur in one or more of the following forms:

(a) Free ammonia, called ammonia nitrogen

(b) Albuminoid nitrogen, called Organic nitrogen

(c) Nitrites

(d) Nitrates

- The free ammonia indicates the very first stage of decomposition of organic matter (thus indicating recently, staled sewage)
- Albuminoid nitrogen indicates quantity of nitrogen present in sewage before the decomposition of organic matter is started
- Nitrites indicate the presence of partly decomposed (not fully oxidized) organic matter
- Nitrates indicate the presence of fully oxidized organic matter

5. Presence of fats, greases, and oils

- Greases, fats and oils are derived in sewage from:
- discharges of animals and vegetable matter
- from the industries like garages, kitchens of hotels and restaurants, etc
- Such matter form scum on the top of the sedimentation tanks and clog the voids of the filtering media

6. Dissolved Oxygen (D.O.)

- It is necessary to ensure at least 4 ppm of D.O. in it; as otherwise, fish are likely to be killed, creating nuisance near the vicinity of disposal
- To ensure this, D.O. tests are performed during sewage disposal treatment processes
- DO is determined by Winkler's method

11. Bio-Chemical Oxygen Demand (B.O.D)

Organic matter is of two types

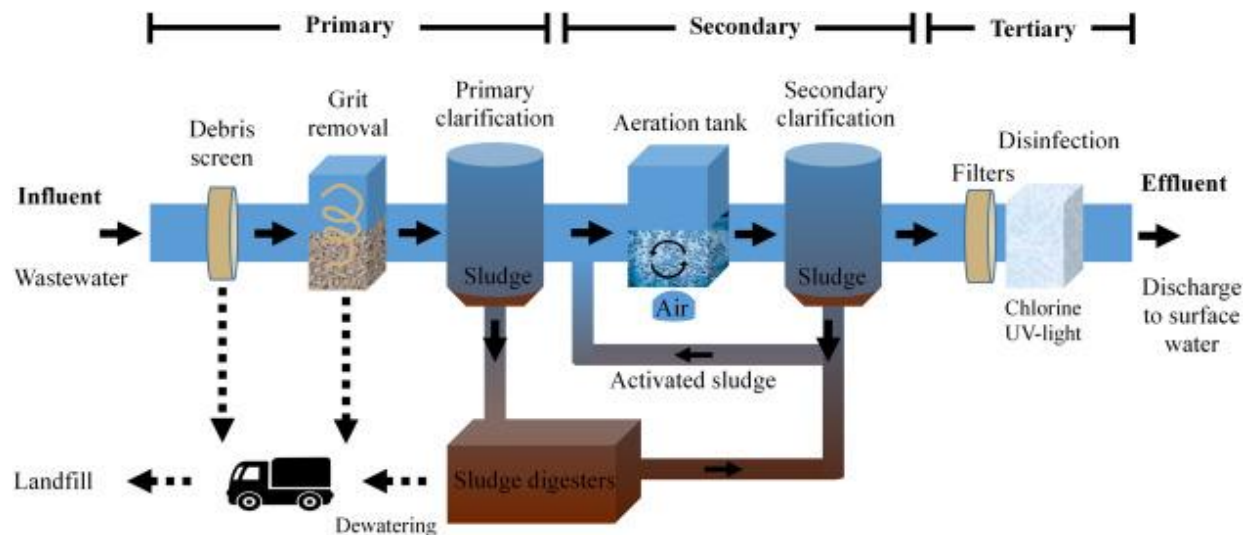
- Biologically oxidized (i.e. oxidized by bacteria) and is called biologically active or biologically degradable
- Organic matter which cannot be oxidized biologically, and is called biologically inactive

8.COD

- Chemical oxygen demand (COD) is a measure of how much oxygen is needed to break down organic and inorganic compounds in water

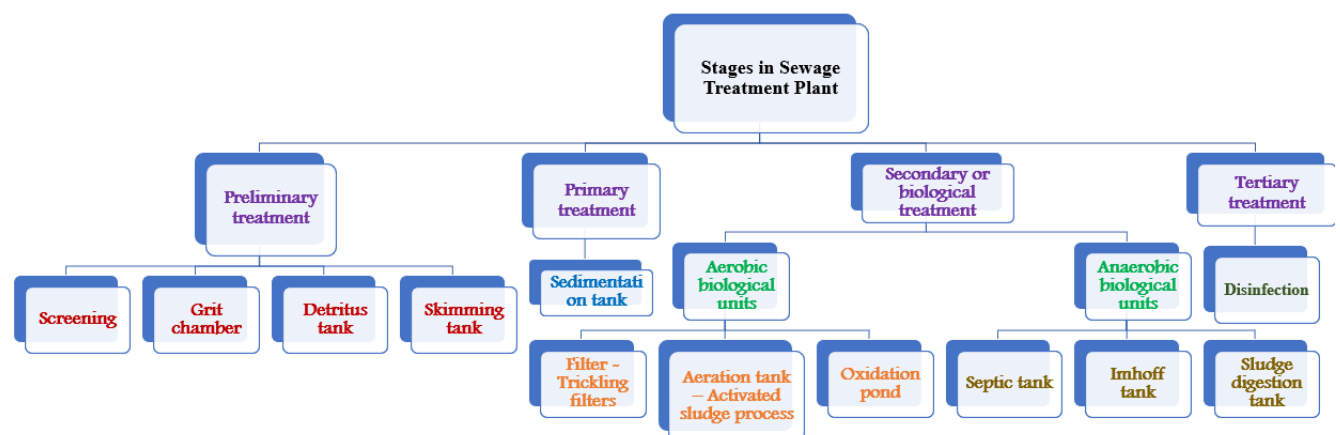
APPLICABLE WASTEWATER DISCHARGE STANDARDS – Refer - The Environment (Protection) Rules, 1986

TYPICAL FLOW SCHEMES FOR SEWAGE TREATMENT PLANT



The sewage treatment units can be broadly classified as

- Primary treatment
- Secondary treatment
- Final treatment



a) PRELIMINARY TREATMENT

1. SCREENING

The main purpose of the installation of screens is to remove the floating matter of comparatively large size (Examples – plants, stones, animals, trees, etc.) to prevent the possible damage of pumps and other equipment's. Types of screens racks or bar-screens, Perforated or

fine screens, Comminutors or cutting screens. The screens should preferably be located just before grit chambers at an angle of 30° to 60° with the direction of flow. The screens are sometimes accommodated in the body of grit chambers. The screening element may consist of parallel bars, rods, gratings or wire meshes or perforated plates and openings may be of any shape; generally, they are circular or rectangular. Types of screens based on spacing between the bars – Coarse, medium & fine screens

- ❖ Coarse screen - > 50 mm
- ❖ Medium screen – 6-40 mm
- ❖ Fine screen – 3-6 mm

2. GRIT CHAMBER

The grit chamber is a settling tank. The detention time for grit chamber is 1 minute. It is used to remove **grit, gravel, sand, egg shells**, etc.

3. DETRITUS TANK

The detritus tank works like a sedimentation tank. The primary clarifier is located after screens and grit chambers. It is used to **remove finer particles** present in wastewater. It is used to remove inorganic suspended solids or grit if it is not removed in grit chamber. The detention time for the detritus tank is 3 to 4 minutes.

4. SKIMMING TANKS

These tanks are used to remove oil, grease, soap, etc. Air diffusers are provided at the bottom of the tank for efficient working of skimming tank. The period of aeration and quantity of air will depend upon the quality of sewage. The compressed air sets up the currents and it results in the floating matter of sewage

- b) **PRIMARY TREATMENT:** In primary treatment, the larger solids from the sewage are removed during the treatment process. The more complex compounds are broken up and converted into simpler compounds by decomposition. The sedimentation tanks with or without use of chemicals.

c) SECONDARY TREATMENT

The effluent that is coming out from primary clarifies contains 45 to 50 percent of the unstable or organic matter originally present in the sewage as solution or suspension or colloidal matter. The sewage to this extent is prepared to receive the secondary treatment. The main function of the secondary treatment of sewage is to convert the remaining organic matter into stable form by oxidation or nitrification.

The secondary or **biological treatment (Aerobic biological units)** involves the following methods:

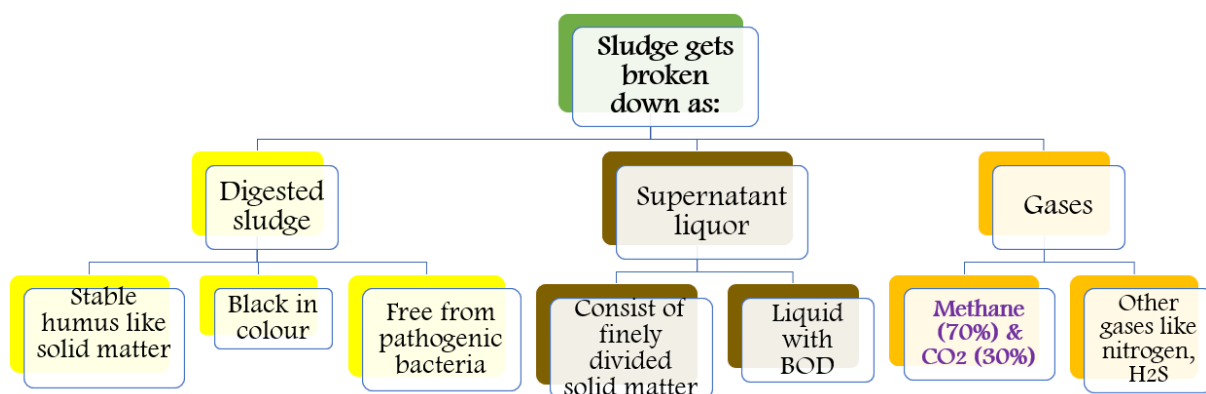
- (i) **Filtration** - Works on the principle of an **attached culture system**. A filter bed consists of highly permeable media to which micro-organisms are attached, through which wastewater is passed. The filters which are commonly employed in the

secondary treatment of sewage are of following types: Contact beds, intermittent sand filters and trickling filters.

- (ii) **Activated sludge process** - Works on **the principle of suspended culture system**. Settled **sludge** in sedimentation tank is called **activated sludge**. **Activated sludge is mixed with raw sewage** along with large quantity of air. Done for about 4 to 8 hours in an aeration tank.

Anaerobic biological units - Sludge digestion tank in the secondary or biological treatment

- Sludge digestion is done to reduce the strength of sludge



d) TERTIARY TREATMENT

DISINFECTION

The filtered water may normally contain some harmful disease-producing bacteria in it. These bacteria must be killed to make the natural water or land area safe. The process of killing these microorganisms is known as disinfection. If pathogenic organisms are present in the treated wastewater, this process needs to be done.

Disposal of Sewage effluent

1. Disposal on water body (dilution)
 - a) In the river
 - b) In the lake water
 - c) In the sea water
2. Disposal on land

1. Disposal of Wastewater by dilution

- Sewage treatment by dilution is the process of discharging treated sewage into a body of water, such as a river, lake, or ocean

- The sewage is diluted into the water's large volume, which eventually purifies the sewage through the natural processes known as the self-purification process of natural waters

Conditions Favouring Disposal by Dilution

- When sewage is comparatively fresh (4 to 5 hours old), and free from floating and settleable solids, (or are easily removed by primary treatment).
- When the diluting water (i.e., the source of disposal) has a high dissolved oxygen (D.O.) content.
- Where diluting waters are not used for navigation or water supply for at least some reasonable distance downstream from the point of sewage disposal.
- Where the flow currents of the diluting waters are favorable, causing no deposition, nuisance, or destruction of aquatic life. It means that swift forward currents are helpful, as they easily carry away the sewage to the points of unlimited dilution. On the other hand, slow back currents tend to cause sedimentation, resulting in large sludge deposits.

1a) Dilution in River and Self-Purification of Natural Streams

The various natural forces of purification which help in effecting self-purification process are summarized below:

1. Physical forces are :

- Dilution and dispersion:** The process by which a large volume of water reduces the concentration of pollutants, and currents spread them over a wider area.
- Sedimentation:** Heavy suspended particles settle to the bottom of the water body due to gravity, effectively removing them from the flowing water.
- Sunlight** (acts through biochemical reactions): Acts as a disinfectant and powers **photosynthesis** in algae, which provides the oxygen necessary for biochemical reactions.

2. Chemical forces aided by biological forces (called biochemical forces) are:

- Oxidation (Bio-oxidation):** Microorganisms use dissolved oxygen to break down organic matter into simpler, stable substances like CO₂ and water.
- Reduction** - Occurs in anaerobic conditions (where oxygen is absent), where bacteria break down organic matter by extracting oxygen from the compounds themselves

OXYGEN SAG CURVE - Zone of Pollution in a River-Stream

A polluted stream undergoing **self-purification** can be divided into the following four zones:

- Zone of degradation
- Zone of active decomposition
- Zone of recovery

(iv) Zone of cleaner water

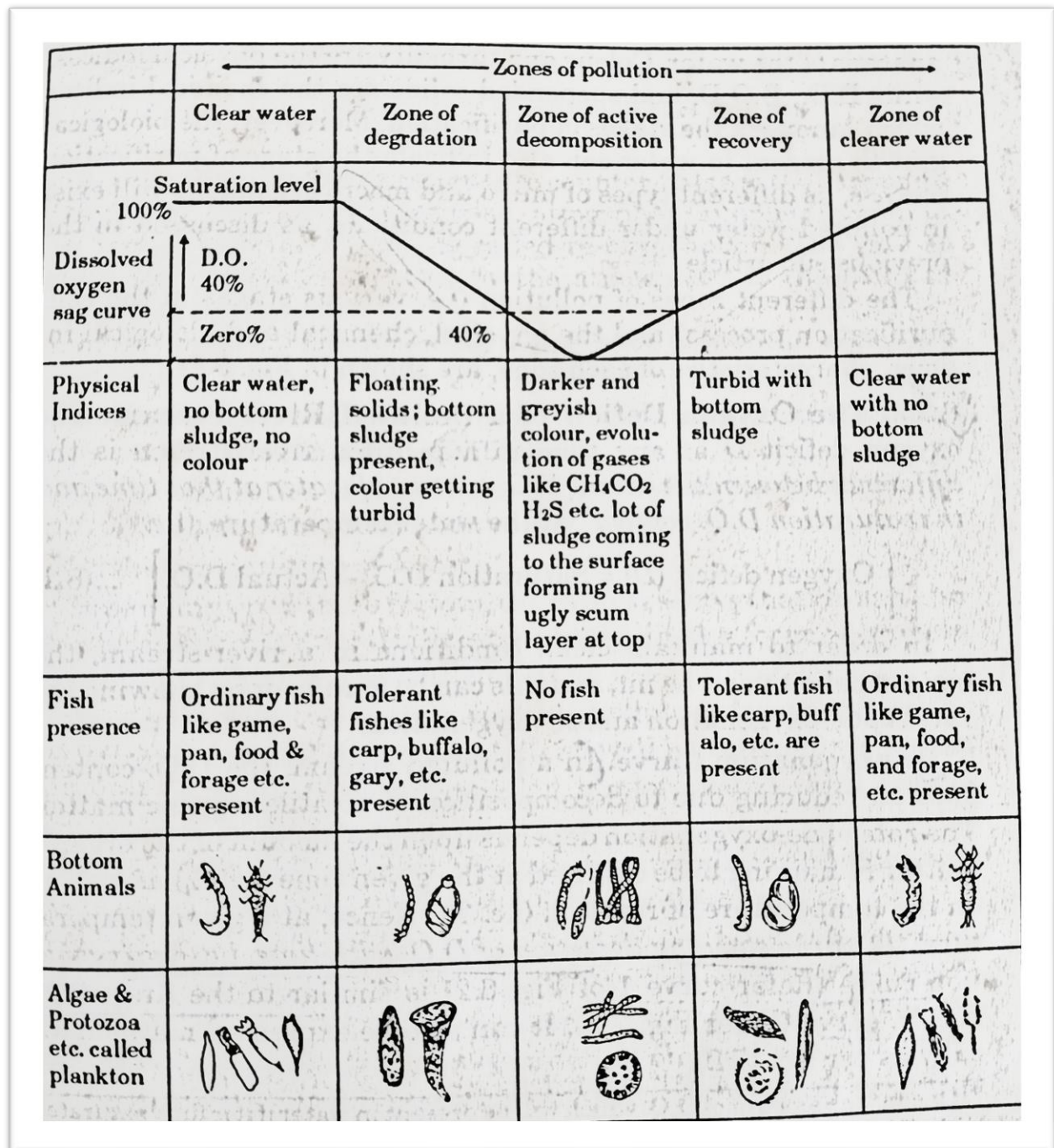


Fig. Oxygen sag curve

(i) Zone of degradation

The zone of degradation is also known as the zone of pollution. This zone is found for a certain length just below the point where sewage is discharged into the river-stream. This zone is characterized by water becoming dark and turbid with the formation of sludge

deposits at the bottom. **D.O. is reduced to about 40% of the saturation value.** There is an **increase in carbon dioxide** content, reoxygenation (i.e., re-aeration) occurs, but is slower than de-oxygenation. These conditions are **unfavorable to the development of aquatic life, and as such, algae die out, but fish life may be present**, feeding on fresh organic matter.

(ii) Zone of active decomposition

The zone of active decomposition is marked by heavy pollution. It is characterized by water **becoming greyish and darker than in the previous zone. D.O. concentration falls to zero**, and anaerobic conditions may set in with the evolution of gases like methane, carbon dioxide, hydrogen sulphide, etc., bubbling to the surface, with masses of sludge forming an ugly scum layer at the surface. As the organic decomposition slackens due to stabilization of organic matter, the re-aeration sets in and D.O. again rises to the original level (i.e., about 40%). In this zone, bacterial flora will flourish. At the upper end, anaerobic bacteria will replace aerobic bacteria, while at the lower end, the position will be reversed. Protozoa and fungi will first disappear and then reappear. Fish life will be absent.

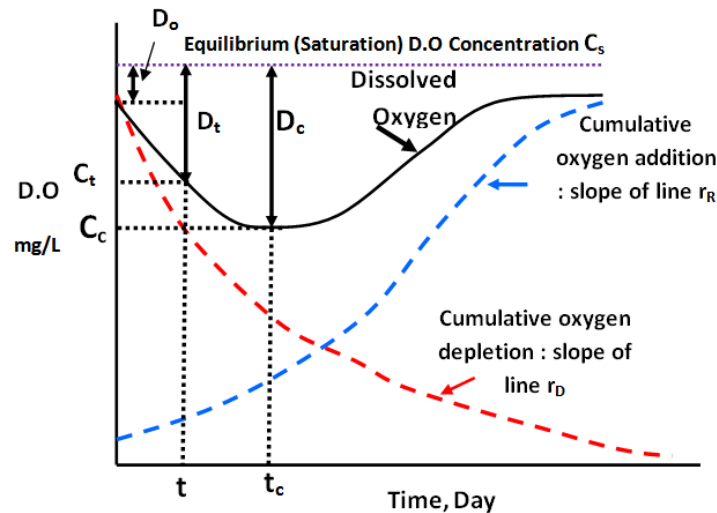
(ii) Zone of recovery

In this zone, the river stream **tries to recover from its degraded condition** to its former appearance. The water becomes **clearer**, and so the algae reappear while the fungi decrease. **B.O.D. falls down, and D.O. content rises above 40%** of the saturation value. Protozoa, and large plants like Sponges, Bryozoas, etc., also reappear. The organic material will be mineralized to form nitrates, sulphates, phosphates, carbonates, etc.

(iv) Zone of cleaner water

In this zone, the **river attains its original conditions with D.O. rising to the saturation value.** Water becomes attractive, and Game fish (which require at least 4 to 5 mg/l of D.O.) and usual aquatic life prevail. Some pathogenic organisms may still, however, survive and remain present, which confirms the fact that "when once a river water has been polluted, it will not be safe to drink it, unless it is properly treated."

De-oxygenation & Re-oxygenation Curve



- **Curve I – Deoxygenation curve or curve showing depletion of D.O. in the absence of aeration**
- **Curve II – Re-oxygen curve (DO content supplied)**
- **Curve III – Oxygen sag curve or oxygen deficit curve**

De-oxygenation Curve

- In a polluted stream, the D.O. content goes on reducing due to the decomposition of volatile organic matter.
- The rate of de-oxygenation depends upon the amount of the organic matter remaining to be oxidized at the given time, as well as on the temperature of reaction.
- Hence, at a given temperature, the curve showing depletion of D.O. with time, i.e. deoxygenation curve. (Curve I)

Re-oxygenation Curve

- To counter-balance the consumption of D.O. due to de-oxygenation, atmosphere supplies oxygen to the water, and the process is called re-oxygenation.
- The rate at which oxygen is supplied by the atmosphere to the polluted water depends upon:

- (i) the depth of the receiving water (the rate is higher in a shallow depth)
- (ii) the condition of the body of water (the rate is higher in a running stream than in a pond)
- (iii) the saturation deficit or the oxygen deficit (i.e., the deficit of D.O. below the saturation value)

- (iv) the temperature of water.

- Depending upon these factors, the rate of re-oxygenation can also be expressed mathematically and plotted in the form of a curve called the re-oxygenation curve (Curve II)

1b) Disposal of Wastewater in the lake water

- Disposal of wastewater in confined in much more harmful than its disposal in flowing streams and rivers
- A river is a flowing water body, while a lake has stagnant waters, so in lakes, only the top surface would become saturated with DO, but the bottom layers would not have enough oxygen
- Overturning of layers would not occur frequently, so that the DO content would not be uniform throughout the depth of the lake
- Overturning takes place only when there is a change in the season, due to which there will be a temperature difference between water in different layers, which cause change in the densities of different layers, and overturning occurs

1c) Disposal of Wastewater in seawater

- Sea water normally contains 20% less oxygen than that contained in the fresh water of a river stream.
- Moreover, seawater normally contains a large amount of dissolved matter.
- As such, the capacity of seawater to absorb sewage solids is not as high as that of the fresh water of a stream.
- Moreover, sewage solids, when thrown into seawater, chemically react with the dissolved matter of seawater, resulting in precipitating some of the sewage solids, giving a milky appearance to seawater and forming sludge banks.
- These sludge banks are undesirable, as they are likely to produce offensive hydrogen sulphide gas by reacting with the sulphate-rich water of the sea.

2. Disposal of Sewage Effluents on Land for Irrigation

In this method, the sewage effluent (treated or diluted) is generally disposed of by applying it on land. The percolating water may either join the water-table, or is collected below by a system of under-drains. This method can then be used for irrigating crops.

This method, in addition to disposing of the sewage, may help in increasing crop yields (by 33% or so) as the sewage generally contains a lot of fertilising minerals and other elements. However, the sewage effluent before being used as irrigation water, must be made safe. In order to lay down the limiting standards for sewage effluents, and the degree of treatment required, it is necessary to study as to what happens when sewage is used for irrigation water, applied on to the land as irrigation water.

When raw or partly treated sewage is applied on to the land, a part of it evaporates, and the remaining portion percolates through the ground soil. While percolating through the soil, the suspended particles present in the sewage are caught in the soil voids. If proper aeration of these voids is maintained, the organic sewage solids caught in these voids get oxidised by the aerobic process. Such aeration and aerobic conditions will more likely prevail if the soil is sufficiently porous and permeable (such as sands and porous loams). However, if the land is

made up of heavy, sticky and fine-grained materials (such as clay, rock, etc.), the void spaces will soon get choked up, and thus resulting in non-aeration of these voids. This will lead to the developing of non-aerobic decomposition of organic matter, and evolution of foul gases. Moreover, excessive clogging may also result in ugly ponding of sewage over the farm land, where mosquitoes may breed in large number, causing further nuisance.

DECENTRALIZED WASTEWATER TREATMENT

- Decentralized systems are small, individual or cluster-type wastewater facilities
 - Done to provide wastewater treatment services to residents
 - Wastewater can be treated onsite through aerobic and anaerobic techniques
 - Decentralized wastewater management (DWWM) systems treat wastewater of individual houses, or apartment blocks, or small communities close to their origin.
 - A decentralized system is a combination of many technologies, namely onsite systems, low-cost collection systems and dispersed siting of treatment.
 - Decentralized Wastewater treatment systems include septic tanks, pit latrines, DEWATS etc.
- ❖ These systems are used for partially treating wastewater in individual residences or a small cluster of houses, are termed as “**On Site Wastewater Treatment (OSWT)**” systems.

1. DEWATS

Anaerobic modules comprise of:

1. Settlers

- Primary treatment, which includes pre-treatment and sedimentation in settlement tank or septic tank
- Also known as a sedimentation basin, settling tank, or clarifier
- Device which **separates** the liquid from the solid
- Retention time is only 2 hours
- Pollution reduction is around 30%

2. Baffle reactors

- A device with several identical chambers
- Through the chamber the effluent moves from top to bottom
- Retention time is 24 hours
- Pollution reduction is around 80%

3. Anaerobic filters

- A device filled with a filter material (cinder) through Which the effluent moves from top to bottom
- Retention time is around 8 hours
- Total pollution reduction is around 90%
- At this stage CPCB (Central pollution control board) standards are met but the effluent still has an odour

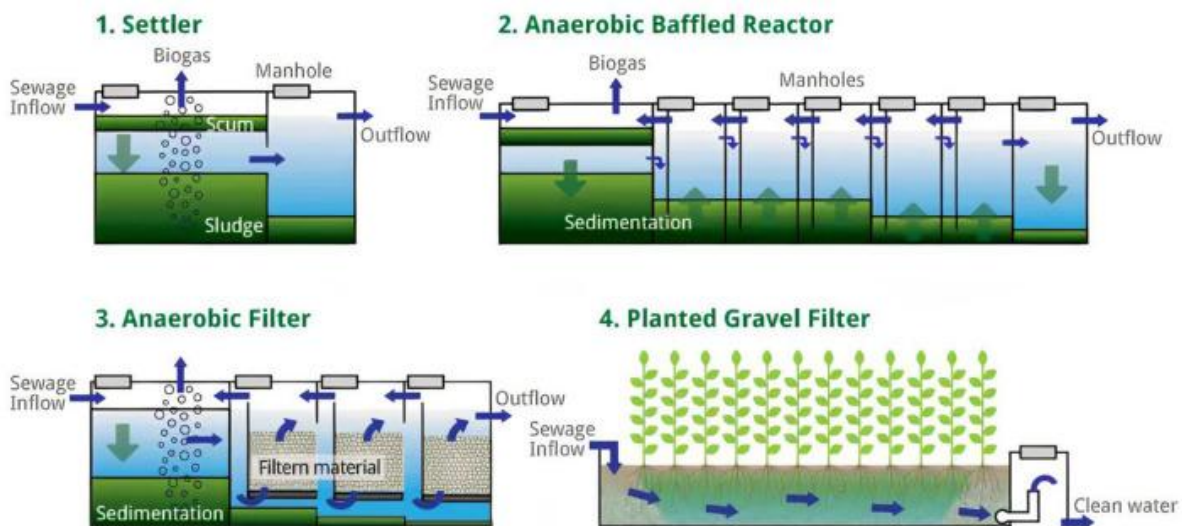
Aerobic modules comprise of:

1. Horizontal planted gravel filters

- Structure filled with gravel material
- Planted with water resistant plants
- Plant provide oxygen to the passing effluent
- Retention time is 1½ day
- Pollution reduction is around 90%

2. Polishing ponds

- Polishing pond is the final stage of treatment
- Acting as an open, shallow water body
- Partially treated wastewater undergoes further biological purification through natural processes like sunlight exposure
- It gives a high-quality effluent suitable for reuse or discharge into the environment
- “Polishing” the water to a clean state before final disposal
- It usually comes after a planted gravel filter in the DEWATS system



2. SEPTIC TANK

- Septic tank, is a horizontal continuous flow type of a sedimentation tank

- It directly admits raw sewage
- It works on the principle of anaerobic decomposition
- Detention period 12 to 36 hours
- Due to anaerobic decomposition of the settled sludge, foul gases will be evolved in the tank
- Septic tank will be a completely covered tank, provided with a high vent shaft for escape of gases
- It removes about 60 to 70% of the dissolved matter
- Effluent from such a tank will be sufficiently foul in nature
- It has to be disposed of either for sub-surface irrigation or in cess-pools or soak pits, or to be treated in trickling filters
- The sludge settled at the bottom of the tank, oils and greasy matter rising to the top surface of sewage as scum, are allowed to remain in the tank for a period of several months
- They are decomposed by the anaerobic bacteria to form gases and liquids by the process of sludge digestion



Advantages of Decentralized Wastewater treatment

- Flows at any point in the system would remain small, implying less environmental damage from any mishap
- System construction results in less environmental disturbances as smaller pipes would be installed at shallow depths and could be more flexibly routed
- The system expansion is easier, new treatment centers can be added without routing ever more flows to existing centers
- Entry of industrial waste could be more easily monitored
- Sector wise treatment permits sewage transmission over shorter distances
- Treatment units are close knit and are free from odours and insects
- Lesser investment is required for the sewer pipelines

- Quality of treatment is more efficient than traditional system due to accurate estimation of wastewater generation and lower quantity of wastewater
- Treated sewage can be effectively used within the sector for applications like toilet flushing, landscape irrigation and cooling tower make ups
- Maintenance of the sewerage system is easier

Dis-advantages of Decentralized Wastewater treatment

- Policies regarding installation, operation and maintenance are not yet well established in many of the developing countries
- Standardization of the systems is difficult as significant variation exists with regard to technical design to suit the local geography and climatic conditions

NATURAL METHODS OF WASTEWATER TREATMENT

1. CONSTRUCTED WETLANDS

Processes involved are wetland vegetation and soils for treating wastewater. Constructed wetlands consist of the following:

- Impermeable liner
- Distribution media
- Plants (Macrophytes)
- Under drainage system

1. Liner is used to keep the wastewater inside the system so that groundwater remains uncontaminated

2. Distribution media is usually crushed rock, which spreads the wastewater across the width of the wetland

3. Locally adapted species of plants that can grow at optimum efficiency are used such as cattails, bulrushes, reeds, and sedges

4. An under-drain system can simply be a slotted pipe covered with distribution media and moves the treated effluent out of the wetland

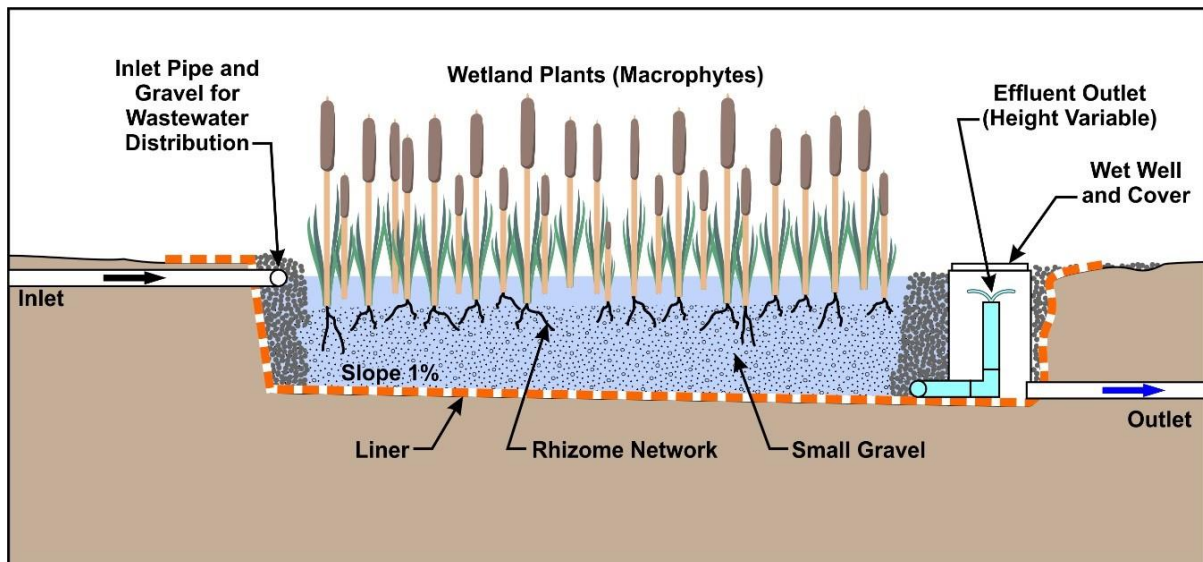


Fig. Constructed Wetland

Constructed Wetlands (CWs) are classified into:

1. **Free water surface flow CWs** - Designed to simulate natural wetlands, with the water flowing over the soil surface at shallow depths
2. **Surface flow CWs** - Designed to create subsurface flow through a permeable medium, keeping the water being treated below the surface. Helping to avoid the development of odours and other nuisance problems

2. LAGOONING

- Lagoons include ponds, like bodies of water or basins
- Designed to receive hold and treat wastewater for a predetermined period of time
- Lagooning is a natural and energy-efficient method for treating wastewater
- The method involves in the accumulation of wastewater in ponds or basins, known as biological or stabilization ponds
- The depth of ponds ranges from 1 to 5 m
- Wastewater stays in the ponds **from two to ten days** in order to be purified



Advantages

- Lagooning is a very **effective and low-cost solution** because of both the initial implementation and the low maintenance
- Lagoon purification is a very efficient method for wastewater treatment to **purify wastewater from small or medium communities** and from communities with variable populations.
- The effluents from lagoon systems are suitable **for irrigation purposes** due to its high nutrient and low pathogen content

3. AERATED LAGOON

- Aerated lagoons are relatively shallow lagoons
- In this, wastewater is added at a **single point, either at the edge or the middle of the lagoon**
- Effluent is removed from another point
- They use aerators to mix the contents of the pond and add oxygen to the wastewater
- Retention time may vary from 6 to 18 days
- Removal of BOD from domestic wastewater varies from 75 to 90%
- Oxygen is supplied by means of surface aerators or by diffused aeration units
- Lagoons may be operated as either aerobic or as aerobic-anaerobic systems
- In aerobic lagoons all biological solids are in continual suspension and stabilization of the organics occurs under aerobic conditions
- Aeration increases the efficiency of treatment and requires only shorter detention period



4. Oxidation ponds

- An oxidation pond is a secondary treatment system
- **Wastewater treated using the principle of algal bacterial symbiosis**
- They treat partially treated wastewater in earthen basins with a depth of about 1-1.5 m
- The bacteria will feed on the waste utilizing the oxygen generated by algae during photosynthesis
- The combined action of bacteria and algae helps to effectively treat the wastewater
- The detention period for oxidation pond is 10 to 30 days.

AEROBIC & ANAEROBIC PONDS

